

AI Provenance and the Evaluation of Intellectual Work: Rational Risk, Social Meaning, and Human–AI Performance

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September 2026

Abstract

Generative AI can change both the quality of intellectual work and how that work is judged once AI use is known. Research on human–AI collaboration mainly asks what AI does to performance; disclosure research asks what evaluators infer from knowing that AI was involved. This paper connects those questions for intellectual work. It argues that the right performance benchmark depends on what the evaluation is trying to measure, and that a binary “AI was used” label often provides too little information by itself. Evaluators may care instead about what function AI performed, how much decision authority the human retained, what was verified, and what or whom is being evaluated. Negative reactions to AI provenance may reflect statistical risk, institutional purpose, social meaning, or mixtures of these explanations. Stigma is therefore treated as a possible interpretation to be tested, not as a starting assumption. The framework also recognizes that expected disclosure can change production behavior before evaluation occurs and that the meaning of AI provenance can change as technology and institutions evolve.

Keywords: AI provenance; AI disclosure; human–AI performance; evaluative objective; benchmark problem; signaling; stigma

1. Introduction

Generative AI is becoming part of the production process for intellectual work. Researchers use it to search, summarize, code, translate, revise language, develop counterarguments, and sometimes generate substantive text or ideas. Comparable uses occur in journalism, consulting, education, hiring, and professional analysis. As these practices spread, institutions increasingly ask whether AI was used and, in some settings, require disclosure.

Consider a simple case. A researcher uses AI while preparing a manuscript and then submits it to a journal that requires AI-use disclosure. AI assistance may have changed the manuscript itself. Separately, learning that AI was involved may change how an editor judges the manuscript, the researcher, or the production process. These possibilities raise two related but distinct questions: Did AI make the manuscript better or worse? And what does knowledge of AI use add to the evaluation?

One literature examines what evaluators infer when they learn that AI was used. Such knowledge can be relevant because generative systems can hallucinate, fabricate references, mishandle confidential information, or fail on tasks outside their capabilities; AI use can also create uncertainty about human contribution, accountability, and rule compliance. But provenance knowledge can affect evaluation in

ways that extend beyond these technical concerns. Experimental studies report lower trust, legitimacy, authenticity, quality, or ethicality in some settings when recipients learn that AI contributed to a work [1–3], while other studies find null or conditional effects [4].

A separate literature shows that AI can change actual performance. In a field experiment with professional consultants, AI improved speed and quality on tasks within the model’s capability frontier but reduced correctness on a task outside it [5]. A broader meta-analysis likewise finds substantial augmentation relative to humans working alone while showing that, on average across the studies examined, human–AI combinations do not outperform the strongest human or AI alternative; effects vary substantially by task and workflow [6].

Taken together, the two literatures reveal a shared evaluation problem: AI can change the work itself, while knowledge of AI use can separately change how that work is judged. Earlier work on algorithm aversion and algorithm appreciation already showed that source knowledge can affect evaluation independently of objective performance [7,8]. This paper connects that insight to hybrid human–AI intellectual production, where AI can change the artifact, the producer’s effective performance, and what the artifact signals about the producer.

Separating what AI does to the work from what knowledge of AI use does to evaluation matters because a negative provenance response is not automatically stigma. Provenance may update beliefs about error risk, verification burden, task–AI fit, human contribution, compliance, or institutional cost. Disclosure can also activate judgments about effort, authenticity, legitimacy, and deservingness that may extend beyond the information relevant to the decision [1,2,9].

The paper therefore asks a narrower question than whether AI-assisted work should be trusted or distrusted. What information should the fact of AI use legitimately add to an evaluation, and how should provenance-based treatment be interpreted when it does not track the attributes the evaluator is actually trying to assess? Stigma is one possible explanation for such treatment, not a premise of the analysis.

The argument proceeds in four steps. First, it separates what AI does to performance from what AI provenance does to evaluation, while allowing disclosure expectations to feed back into production. Second, it shows that the right benchmark depends on the purpose of the evaluation rather than on one universal human-versus-AI comparison. Third, it treats statistical risk, institutional purpose, and social meaning as competing and sometimes overlapping explanations for provenance effects. Fourth, it asks how those effects may change as AI, norms, and institutions evolve.

2. Performance, Provenance, and the Benchmark Problem

2.1 Performance and provenance are different questions

The human–AI performance literature asks what happens to work when AI is added to the production system. The provenance literature asks what happens to evaluation when recipients know or believe that AI was involved. These are different causal questions, even though they can interact in practice.

Let ΔQ represent the change in performance attributable to AI assistance under a specified task and workflow. In an augmentation setting, a useful conceptual benchmark is:

$$\Delta Q_i = Q(H_i + AI) - Q(H_i)$$

This quantity asks whether AI improved the work of producer i relative to that producer's unaided counterfactual. It does not ask whether AI is superior to an expert human or whether the combined system is superior to the best available alternative.

Different performance benchmarks can yield different answers. AI may improve a particular producer relative to that producer's unaided baseline even when an expert evaluator outperforms the AI system or the resulting work remains below a field standard. The BCG field experiment and the broader meta-analysis show why benchmark choice matters: augmentation can be positive relative to humans alone while failing to create superiority relative to the better standalone alternative [5,6].

2.2 The benchmark follows from the evaluative objective

The individual unaided counterfactual is therefore important but not universal. In a professional setting concerned with producing the best reliable report, the relevant question may be whether AI improves the feasible production system. In a licensing exam or writing assessment, the purpose may be to measure unaided human capability. In hiring or competitive allocation, relative ranking among candidates can be central. In journal review, the relevant standard may be whether the work advances the field rather than whether the author improved relative to personal baseline.

A benchmark appropriate to one evaluative purpose can be misleading when imported into another without justification. Evaluating whether AI should replace expert judgment is different from evaluating whether AI augmented a particular producer. Evaluating an artifact is different from credentialing the producer who created it.

The relevant counterfactual should be derived from the evaluative objective; no single human-versus-AI benchmark is universally appropriate.

2.3 Four performance–evaluation configurations

Let ΔE denote the evaluation effect associated with known AI provenance. Performance and evaluation can therefore move together or in opposite directions:

Table 1. Performance and provenance-evaluation configurations

Performance effect	Provenance evaluation effect	Interpretation
Positive	Positive	Beneficial assistance is positively recognized
Positive	Negative	AI improves the work but provenance reduces evaluation
Negative	Negative	AI harms the work and provenance appropriately signals concern
Negative	Positive	AI harms the work but provenance produces unwarranted confidence

The second case motivates concern about provenance-based devaluation because a productivity-enhancing intervention can be penalized. The fourth case provides the symmetric warning: positive social meaning attached to AI can also misprice performance. The framework is therefore about alignment between provenance-based evaluation and the attributes the decision is meant to assess, not about presuming anti-AI bias.

The four configurations must also be interpreted relative to the benchmark each evaluator is using. A researcher may reasonably focus on the incremental improvement created by AI, while an editor focuses on whether the resulting paper crosses a field-level threshold. Both judgments can be internally coherent because they answer different questions. Problems arise when the grounds for one judgment are used to support another—for example, when an editor’s belief that an AI system is inferior to expert judgment is taken as evidence that the system probably made a particular author worse. The framework therefore separates benchmark selection from benchmark performance: first specify the objective, then ask how AI affected performance relative to the benchmark that follows from that objective.

Specifying the evaluative objective before interpreting performance also avoids a productivity-first assumption. A positive ΔQ is not automatically a social gain. If AI improves an examination answer while defeating the purpose of measuring unaided reasoning, the improvement is irrelevant or adverse to the institution’s objective. If AI helps an analyst produce a more accurate forecast under conditions where AI will also be available in practice, the same improvement can be directly valuable. The framework therefore treats productivity, credential signaling, and competitive allocation as distinct evaluative logics rather than ranking one above the others.

Benchmark choice determines the standard against which performance is judged; production architecture determines what kind of human–AI system produced the artifact. AI can substitute for a human function, augment it, or change the division of labor inside the production system. The same observed artifact may result from substantially different combinations of human judgment, machine generation, review, and revision. These differences help explain why provenance can matter to evaluators even when observed outputs are similar.

3. AI Provenance as Cue, Signal, and Target

3.1 Provenance is a cue, not the attribute itself

A binary statement such as “AI was used” is informationally sparse. It does not reveal what function AI performed, which system was used, how much authority the human retained, whether output was verified, whether the task lay within the model’s capability frontier, or whether the resulting work improved relative to the relevant counterfactual.

Recipients nevertheless draw inferences from the cue. They may infer higher error risk, greater verification burden, weaker human oversight, reduced intellectual contribution, lower effort, questionable compliance, or reduced authenticity and legitimacy [1–3,9]. Some of these inferences may be decision-

relevant; others may be weak proxies or socially constructed meanings. Treating them as a single “AI attitude” obscures the mechanism.

3.2 What is being evaluated?

The same provenance cue can affect different targets. When the target is the artifact, relevant attributes include accuracy, reasoning quality, originality, and usefulness. When the target is the producer, the evaluator may care about competence, independent reasoning, authorship, contribution, or trustworthiness. When the target is the process, compliance, transparency, reproducibility, or retained human authority may matter. Institutions themselves can also be evaluated for how they permit or rely on AI.

This target distinction explains why identical artifacts need not receive identical judgments in every setting. A paper can be technically sound while providing weaker evidence of an author’s unaided capability. A résumé can accurately describe a candidate while revealing less about unaided writing skill. A student answer can be correct while failing the purpose of an exercise designed to measure independent reasoning.

3.3 Signaling and credentialing

In many markets and institutions, a work product matters for two reasons: it has value in itself, and it can reveal something about the person who produced it. A writing sample, examination answer, or research paper may therefore be used as evidence of an underlying capability that is costly to observe directly [13]. Signaling theory provides a useful language for this second role.

AI can improve the visible artifact while changing what that artifact reveals about the producer. If assistance raises the observable performance of weaker producers more than stronger producers, work from people with different unaided capabilities may become more similar. In an output-oriented setting, that compression can be beneficial because more people can produce strong work. In a credentialing setting, however, it can weaken the artifact’s ability to distinguish latent unaided capability. The same improvement in artifact quality can therefore have different implications depending on whether the institution values the output itself or the information the output carries about the producer.

The signaling implication is that provenance can remain relevant even when the artifact improves. If the institution is buying an output, augmentation may be desirable. If the institution is trying to infer unaided capability, extensive assistance can reduce the diagnostic value of the artifact even when the artifact itself improves. In that setting, provenance matters because the mapping from observed work to latent human capability has changed.

3.4 Function and retained authority matter more than presence alone

Useful disclosure therefore needs to identify both what AI did and how decision authority was allocated. Language correction, coding assistance, literature organization, synthesis, hypothesis generation, substantive drafting, and autonomous decision-making imply different risks and different amounts of

human contribution. Function alone is not enough: AI can generate substantial text while the human retains authority through source selection, rejection, recombination, and verification, or it can generate relatively little while the human accepts output without meaningful scrutiny.

The relevant information is thus not simply whether AI appeared in the workflow, but the function AI performed and the cognitive and decision authority the human retained.

Cue validity can therefore vary sharply across uses. If a disclosure merely states that AI was involved, evaluators may fill the missing information with assumptions based on the most salient examples of AI failure or misuse. One recipient may imagine language polishing; another may infer large-scale autonomous generation. The same label can consequently produce heterogeneous responses even before differences in attitudes toward AI are considered.

More specific disclosure can reduce some of this ambiguity, but specificity is not guaranteed to eliminate provenance effects. A detailed statement that all claims and citations were verified may reduce technical uncertainty while leaving judgments about effort, authorship, or authenticity untouched. Conversely, specifying that AI was used only for language revision may be highly informative in a research-quality evaluation but insufficient in an assessment whose purpose is to measure unaided writing ability. The informational value of disclosure is therefore conditional on function, retained authority, and evaluative target.

Because provenance can change what is being inferred even when the artifact is unchanged, this framework also affects how identical-output studies should be interpreted. Such experiments are valuable because they isolate the causal influence of provenance information on evaluation, but they do not automatically establish irrationality. If the evaluator is judging factual quality alone, identical outputs provide a strong control. If the evaluator is judging authorship, individual capability, or compliance, provenance can change the object of inference even when the artifact is unchanged. Experimental interpretation must therefore specify what attributes have actually been held constant relative to the evaluative objective.

4. From Rational Inference to Residual Provenance Effects

4.1 Three classes of explanation

Negative provenance effects can arise from at least three classes of explanation. A predictive inference uses provenance because it forecasts a decision-relevant attribute such as error or verification burden. A purpose-based institutional inference follows from the objective or rules of the evaluation, such as measuring unaided reasoning in an examination. A social-meaning inference attaches meaning to AI provenance through judgments such as authenticity, effort, legitimacy, or deservingness. These classes can overlap; they are analytical starting points rather than mutually exclusive empirical bins.

Table 2. Three classes of explanation for provenance effects

Explanation	Illustrative example	Evidence that would weaken it
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Explanation	Illustrative example	Evidence that would weaken it
Statistical	Citation or verification risk	Direct verification; smaller observed risk differential
Institutional	Exam measures unaided reasoning	Change in the evaluative objective or rule
Social	Authenticity or deservingness	Risk/contribution controls; mediation by social meaning

The analytical task is therefore not to subtract a “rational” quantity from an observed penalty. It is to test competing explanations against the evaluative objective and available evidence.

4.2 Artifact risk and verification burden

Generative AI introduces distinctive failure modes: hallucinated facts, fabricated references, unsupported synthesis, and plausible but incorrect reasoning. Where AI use empirically predicts such failures, provenance can change an evaluator’s prior about expected quality. Provenance may also matter because some AI-assisted workflows raise the expected cost of checking whether those failures occurred. Error risk and verification burden are related but distinct reasons for scrutiny.

But relevance is not precision. The same binary label can cover low-risk language assistance and high-risk substantive generation. As direct evidence becomes available—verified references, reproducible analysis, explicit human review—the marginal value of generic provenance as a proxy should decline.

4.3 Contribution, accountability, and compliance

AI provenance can also be directly relevant when the evaluation concerns attributes of the producer or process rather than artifact quality alone. Intellectual authorship may require understanding, adoption, and responsibility for claims, making human contribution and accountability relevant. Compliance is a related institutional concern when a task or rule specifies what assistance is permitted.

These institutional concerns become harder to interpret when rules are incomplete or differently applied during technological transition. Such ambiguity can create opportunities for norm arbitrage without necessarily implying misconduct. A governance-based explanation for provenance effects should become more specific as standards stabilize. Current journal policies illustrate why the transition matters: a lifecycle-based study of health-related journals found that 72.6% prohibited AI authorship, while 54.2% provided no guidance on methodological design, and policy rigor varied with policy presentation mode and journal prestige [14].

4.4 Effort, legitimacy, and social meaning

AI can reduce the human effort required to produce polished work. Effort may be intrinsic to the objective, a proxy for quality or commitment, or a symbolic source of value. These roles should not be conflated. In an educational exercise, effort can be part of the product. In outcome-oriented analysis, reduced effort may instead represent productivity.

Experimental work on AI disclosure suggests that legitimacy, authenticity, commitment, and related judgments can mediate negative reactions even when technical performance is not the only concern [1,2,11]. The effort heuristic provides an older psychological basis for why lower inferred production

effort may reduce value in ambiguous settings [9]. These mechanisms can be socially meaningful without being inherently irrational; their relevance depends on what the institution or recipient is legitimately evaluating.

4.5 Progressive control rather than mechanical decomposition

When evaluators react differently after learning that AI was used, the first task is to ask why. Researchers can test the most direct explanations one by one: Was factual verification uncertain? Was the human role unclear? Did the use violate a rule? Did AI actually improve or harm performance? If the provenance penalty falls as these uncertainties are removed, decision-relevant information was part of the effect. If adverse treatment persists, researchers can then examine whether legitimacy, authenticity, effort, status, or other social meanings help explain what remains.

This approach does not produce a single, uniquely measurable “stigma coefficient.” Statistical, institutional, and social mechanisms can interact, and their relevance depends on the purpose of the evaluation. “Residual stigma” is therefore best understood as an interpretation supported by increasingly strong controls and mechanism evidence, not as a number that can simply be subtracted from an observed penalty.

The same evidentiary logic has an institutional implication: provenance is most useful as a proxy when it demonstrably predicts the attribute being assessed and direct evidence is costly or unavailable. Reliance on provenance should weaken when the underlying attribute can be observed more directly. If the concern is citation reliability, validated references are superior evidence. If the concern is individual contribution, process documentation or oral defense may be more informative. If the concern is review burden, observed manuscript characteristics may eventually outperform AI status as a triage variable.

This creates a general progression from proxy to measurement. Early in technological adoption, provenance can be an inexpensive signal under uncertainty. As institutions collect better data and workflows mature, they can replace coarse category-level assumptions with direct indicators. Older proxies may nevertheless persist because of institutional inertia, switching costs, or limited measurement infrastructure even after their predictive value has weakened.

After competing explanations have been tested and more direct evidence has been introduced, stigma becomes a more plausible interpretation of any remaining devaluation. A negative reaction should not be labeled stigma merely because it is uncomfortable for the producer or because it differentiates AI-assisted work. The interpretation becomes stronger when evaluators possess credible information about performance, verification, contribution, and compliance yet continue to devalue the artifact or actor through judgments that attach socially disfavored meaning to AI provenance. Even then, the mechanisms may remain mixed rather than cleanly separable.

The same framework must also accommodate positive provenance effects. AI provenance can produce a halo when recipients associate the technology with sophistication, innovation, or analytical power and consequently overvalue the work. Algorithm appreciation demonstrates that machine provenance can sometimes increase rather than reduce reliance [8]. A general provenance framework must therefore

explain positive, negative, and null effects rather than treating anti-AI devaluation as the only phenomenon of interest.

5. Provenance Regimes Can Change Production

5.1 Anticipated disclosure creates endogeneity

If producers know in advance that AI use must be disclosed—or expect that disclosure will affect how their work is judged—they may change how they work before the artifact even exists. They may prompt differently, verify more or less, conceal use, avoid AI, or redirect it toward functions they believe will be viewed more favorably.

This means the evaluation regime can become part of the production process itself. Recent evidence on the indirect disclosure effect shows that expected AI disclosure can change creator behavior, effort, and the resulting output before audience evaluation occurs [12]. In causal terms, anticipated disclosure creates endogeneity: provenance expectations are not merely labels applied after production; they can help shape the work that is later evaluated.

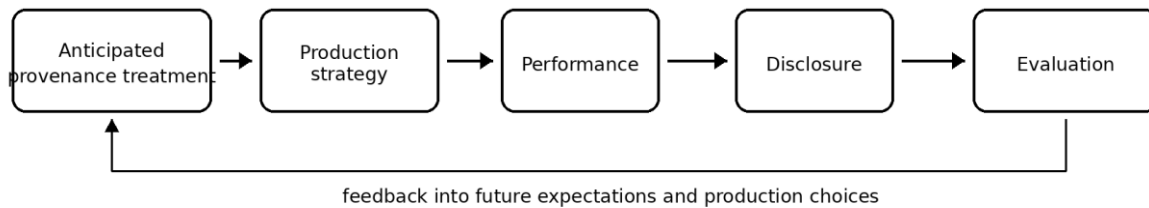


Figure 1. Anticipated disclosure creates an endogenous production–evaluation pathway.

This endogeneity does not erase the analytical distinction between performance and provenance effects, but it means they are causally separable only under specified designs. In observational settings, the provenance regime may already have changed the work being evaluated.

5.2 Producer responses

Once producers expect provenance-based treatment, several strategies become possible: use and disclose; use and conceal; avoid AI despite expected performance gains; or redirect AI toward functions perceived as more acceptable. Preliminary educational evidence links fear of judgment and stigma to concealment [10], and the broader disclosure-paradox literature shows that people can favor disclosure while reacting negatively to disclosed AI use [11].

One especially important possibility is beneficial non-adoption: AI would improve the artifact under the relevant objective, but anticipated provenance costs discourage its use. If unaided performance is the objective, such non-adoption is appropriate. The concern arises when the evaluation system discourages a technology that would improve the very outcome the institution says it wants.

5.3 Institutional triage and proxy use

AI also changes production economics, and several institutional mechanisms can interact. Lower production costs can increase manuscripts, applications, proposals, and other artifacts faster than expert evaluation capacity. Under such scarcity, institutions may use AI provenance as a triage variable when they believe it predicts verification cost, expected quality, or compliance risk. Provenance can also persist as a proxy after its predictive value changes, while producer responses can alter the sample from which institutions learn. These are plausible mechanisms rather than established universal effects; operational evidence is needed to determine when they actually govern review decisions.

The practical distinction is between using provenance as evidence and using it merely as a category. Risk-based triage uses AI status because it predicts something the institution actually cares about, such as verification burden. Category-based triage uses the flag mainly because it is easy to observe. The difference is empirical: researchers should test whether AI status still predicts review treatment after direct measures of quality, verification burden, contribution, and compliance are included.

Because provenance-based triage can be coherent from an institution's perspective even when it disadvantages some producers, "institutional rationality" requires a narrow definition here. It means pursuing the institution's stated evaluative objective coherently under its existing constraints. It does not mean that the objective is normatively neutral or that decision-makers are free of bias. Existing systems embody investments in expertise, reputation, workflows, and allocation mechanisms. The relevant question is whether a provenance-based rule continues to serve its stated purpose as technology, use patterns, and evaluation infrastructure change.

Producer responses can also change the population that institutions observe. If users expecting penalties conceal AI use, disclosed cases may overrepresent highly compliant producers or contexts where disclosure is unavoidable. If weaker producers rely more heavily on AI, the observed association between disclosed use and quality may move in the opposite direction. Institutions then update beliefs from a selected sample, potentially reinforcing or weakening the original provenance rule. Observational comparisons of disclosed and undisclosed work must therefore be interpreted with care.

Resource scarcity also determines whether a provenance flag has any operational value. Lower production cost can increase the volume of manuscripts or applications without proportionately increasing expert evaluation capacity, encouraging institutions to seek inexpensive signals for allocating attention. AI provenance may initially be attractive because it is salient and easy to collect. But a binary flag can reduce substantive review cost only if it changes the review path—through additional verification, specialized routing, deprioritization, or rejection. The operational function of disclosure is therefore an empirical question distinct from its stated rationale.

A further complication arises when AI becomes ordinary on both sides of evaluation. A candidate may use AI to improve a résumé that is screened by AI; an author may disclose AI assistance to a journal that itself relies on automated triage. These arrangements are not necessarily inconsistent because the functions and risks differ, but they show that AI provenance acquires meaning partly through institutional role. As AI becomes ordinary on both sides of evaluation, rules focused only on the producer's technology use may require increasingly specific functional justification.

6. The Evolution of Provenance Value

AI provenance is a time-dependent signal. Its technical information value depends on model reliability, user capability, workflow maturity, verification practices, governance, access, and institutional adaptation. Its social information value depends on legitimacy, authenticity, effort, identity, and status. These components need not evolve together.

Technical-risk effects may fall as reliability and verification improve. Governance-related effects may decline as rules stabilize. Access-based fairness concerns may weaken as tools become widespread. Contribution concerns may persist where individual capability is the object of assessment. Authenticity concerns may remain strong in domains where human origin is part of the product. Institutional congestion may worsen during rapid adoption and then recede as evaluation systems adapt.

The central dynamic prediction is that the composition and usefulness of the provenance cue will change as the conditions that give it meaning change. Some components may weaken while others persist or intensify. Cross-sectional disclosure studies therefore measure a historically situated effect rather than a permanent property of AI-assisted work.

6.1 A limiting case: widespread adoption

Consider a future in which 90 percent of serious intellectual work materially depends on AI assistance. “AI-assisted” would no longer identify an unusual production method. A generic presence label would have little sorting value if nearly every work carried it, and institutions could not efficiently treat most intellectual production as an exceptional category. The evaluative question would shift toward what AI did, what remained under human authority, what was verified, and what the relevant task requires.

Even under widespread adoption, provenance can remain relevant when unaided or human-only production is part of the evaluative objective. Examinations, licensing, competitions, and authenticity-sensitive domains provide clear examples. The absence of a dominant technology may itself become a valuable signal. Widespread adoption therefore changes which provenance information is decision-relevant rather than making provenance uniformly irrelevant.

6.2 From technology presence to evaluation design

As AI use becomes routine, disclosure may need to migrate from technology presence toward function- and risk-specific information. Instead of merely stating that AI was used, an evaluator may need to know whether AI generated unverified factual claims, performed substantive analysis, influenced a decision, or operated under specified human review.

This shift is consistent with the broader principle developed throughout the paper: provenance is useful to the extent that it helps identify attributes relevant to the decision. Its institutional weight should therefore be expected to change as more direct information becomes available.

Normalization can reduce category salience while preserving or creating other distinctions. Existing evidence already shows that provenance effects are context dependent, and social meanings can persist

even when technical beliefs improve [1,2,4]. If AI becomes ordinary infrastructure, human-only production may become the salient exception in some domains, while in others provenance may disappear from routine evaluation altogether.

For this reason, the most useful dynamic object is not the aggregate penalty but its composition. A stable overall effect could conceal declining technical-risk concerns and rising authenticity concerns, or the reverse. Longitudinal and repeated experimental designs should therefore measure mediators and evaluative targets rather than simply asking whether the average disclosure effect increased or decreased.

Technological maturity is also multidimensional. Model reliability can improve while deployment volume increases, creating more total errors despite lower per-use risk. User competence can improve while institutions face greater congestion. Governance can become clearer while social conflict over authorship intensifies. The framework therefore predicts changing relationships among mechanisms, not a monotonic path toward acceptance.

7. Implications and Research Priorities

The five research priorities below address one common empirical problem: whether AI provenance helps evaluators reach the decision they actually intend to make, or whether it becomes a coarse substitute for more direct evidence. They are followed by an integrated research design and the framework's scope conditions.

7.1 Specify the evaluative objective and target

Research and policy should begin by specifying what is being evaluated: artifact quality, producer capability, intellectual contribution, process compliance, or institutional legitimacy. Without this step, apparently contradictory results may simply reflect different objectives or targets.

Benchmark choice follows from the same evaluative objective. Studies should state whether the relevant comparison is the producer's unaided counterfactual, a population ranking, a common field standard, or the best available production system.

7.2 Replace binary presence with functional information

Future disclosure research should distinguish language editing, coding, retrieval, synthesis, hypothesis generation, drafting, and autonomous decision-making. It should also specify retained human authority, verification, and any material residual risk. These details are closer to the attributes evaluators actually care about than a binary AI flag.

Where direct evidence of the relevant attribute is available, it should generally carry more weight than provenance as a proxy. Verified references are more informative than AI status if citation reliability is the concern; a clear account of human contribution is more informative if authorship is the concern.

7.3 Measure performance and provenance together

The central empirical gap remains the joint estimation of performance and provenance effects within the same design. A strong study would randomly assign AI access, measure objective task performance, and then randomize what independent evaluators are told about provenance. Additional conditions could progressively disclose verification, human oversight, contribution, and compliance.

Such designs would identify whether evaluative treatment tracks the actual effect of AI assistance and which competing explanations survive stronger controls. They would also permit the symmetric study of negative stigma and positive AI halo effects.

7.4 Measure endogeneity and behavior

Because anticipated disclosure can change production before evaluation occurs, studies should measure more than ratings. Relevant outcomes include AI adoption, prompting strategy, verification effort, concealment, task choice, submission behavior, reviewer effort, and allocation decisions. Experimental manipulation of disclosure expectations before production is especially important [12].

Institutional data could also reveal whether AI disclosure affects desk rejection, review priority, verification requests, or acceptance after controlling for observable quality and burden. This would help distinguish risk-based from category-based triage.

7.5 Replicate over time

Provenance effects should be replicated as technology and norms evolve. Changes in model reliability, AI literacy, prevalence of use, governance, and institutional adaptation are theoretically relevant moderators, not merely background noise. Longitudinal evidence can show whether current penalties reflect transitional uncertainty, persistent social meaning, or changing mixtures of both.

7.6 Integrating the research program

Taken together, the preceding priorities imply an integrated empirical program that combines production-stage and evaluation-stage randomization. Participants could first be assigned AI access or no access on an intellectual task, allowing direct measurement of performance effects. Evaluators could then receive randomized provenance information, with additional conditions specifying verification, human contribution, or compliance. A further manipulation could inform producers in advance about disclosure requirements, revealing whether the provenance regime changes production behavior before the work is judged. This design would connect the human–AI performance literature with disclosure research while directly testing the endogeneity highlighted by the indirect disclosure effect [12].

Institutional studies can complement laboratory experiments. Journals, universities, and employers increasingly possess operational data on review routes, verification requests, processing time, acceptance, and disclosure. Where accessible, these data can test whether AI provenance predicts real evaluation cost

after controlling for direct measures of quality and complexity. They can also show whether disclosure acts mainly as transparency, risk routing, compliance evidence, or a resource-allocation proxy.

Across both experimental and institutional studies, measurement should preserve the distinction among artifact, actor, process, and institution. A provenance cue may have little effect on perceived artifact accuracy but a large effect on intellectual credit assigned to the author. It may reduce trust in the production process while increasing trust in the institution for being transparent. Collapsing these responses into a single attitude measure can obscure both decision-relevant mechanisms and social meaning.

The integrated program also provides a clearer evidentiary standard for claims about residual stigma. The strongest designs will not merely show that AI-labeled work receives lower ratings. They will identify the evaluative objective, establish what AI actually did to performance, clarify human authority and compliance, and test whether negative treatment persists when the most plausible decision-relevant explanations are weakened. The result may be stigma, decision-relevant caution, mixed mechanisms, or no meaningful provenance effect at all.

7.7 Scope conditions

The framework also has scope conditions. It addresses intellectual work in settings where provenance can plausibly influence evaluation and evaluators retain some discretion; it is not a general model of safety-critical AI deployment. “AI” is treated as a broad production category even though model capabilities and workflows vary substantially. The framework organizes competing explanations and research designs rather than claiming that statistical, institutional, and social mechanisms can always be cleanly separated in observational settings. The literature and policy evidence in this version reflects sources available through September 3, 2026; subsequent developments fall outside this point-in-time evidence base.

8. Conclusion

AI-assisted intellectual work cannot be evaluated soundly by asking only whether AI was used. Human–AI performance and provenance-based evaluation are different questions, but they can become causally intertwined when disclosure expectations change how people work.

Provenance can legitimately inform evaluation when it predicts risk, verification burden, contribution, compliance, or the signal value of an artifact. It can also acquire social meanings concerning effort, authenticity, legitimacy, and deservingness. These mechanisms should be tested as competing explanations rather than forced into a clean numerical decomposition.

The correct benchmark and the relevance of provenance both follow from the evaluative objective. Improvement relative to an individual’s unaided baseline matters in augmentation settings; unaided capability, relative ranking, field standards, or best-achievable output may matter elsewhere. The central mistake is not choosing one particular benchmark, but treating a benchmark appropriate to one purpose as universally decisive.

As intellectual production becomes more hybrid, binary AI labels will often become less informative than the function AI performed, the authority retained by humans, the verification applied, and the residual risks relevant to the decision. The central research question is therefore whether provenance-based treatment tracks what AI actually did and what the evaluation legitimately seeks to determine—or whether the label carries evaluative meaning beyond those attributes.

The practical question is simple even when the evidence is not: when an evaluator learns that AI helped produce a piece of work, what should that fact change? Answering it requires knowing what AI actually did, what the evaluation is meant to measure, and whether the provenance cue adds information beyond more direct evidence about quality, contribution, verification, or compliance. Future studies that measure performance, disclosure effects, and production-stage responses together can show when AI provenance improves evaluation, when it becomes an inefficient proxy, and when adverse treatment is better explained by social devaluation than by decision-relevant caution.

AI Use Disclosure

Generative AI tools were used during the development of this work to support literature discovery, conceptual exploration, drafting, editing, and review. The research questions, analytical framework, interpretation of evidence, selection of arguments, and final judgments remain the responsibility of the author. References and substantive claims were checked against source materials where applicable.

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